

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1383

Scottsville 138 kV
October 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1383. The Interconnection Customer proposes a 382.7 MW solar photovoltaic facility interconnecting at the Scottsville 138 kV substation in the Virginia area. The request was received on January 13, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 392 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,272,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
YANKEE DOODLE / ROCK ISLAND DAM 138kV Line	REACH / BLUFF CITY 138kV	187.9	217	65.8
ROCK ISLAND DAM / SCOTTSVILLE 138kV	REACH / BLUFF CITY SUBSTATION 138kV	182.8	235	65.8
YANKEE DOODLE / RED HILL 138kV	BLUFF CITY / REACH 138kV	138.8	270	65.8
REACH / BLUFF CITY 138kV	SCOTTSVILLE / ROCK ISLAND DAM 138kV line	134.4	397	66.0
CHARLOTTESVILLE / MOHICAN 138kV	RED HILL 345 kV/138 kV Transformer	130.0	168	15.8
AMBER / BREMO BLUFF 138kV	BLUFF CITY / LAKE LURE 138kV	116.9	194	11.6
BARRACKS RD SUBSTATION / MOHICAN 138kV line	RED HILL 345 kV/138 kV TX	116.2	200	15.8
BARRACKS ROAD / TIMKEN MERCY SUBSTATION 138kV Line	RED HILL 345/138kV XFMR	116.2	223	15.8
SCOTTSVILLE / REACH 138kV L	SCOTTSVILLE / ROCK ISLAND DAM 138kV	105.1	345	100.0
WALWORTH / AMBER 138kV	BLUFF CITY / LAKE LURE 138kV	102.2	214	11.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
YANKEE DOODLE / ROCK ISLAND DAM 138kV Line	Rebuild line with higher-capacity conductor	\$8,917,000	99% (\$8,845,000)
ROCK ISLAND DAM / SCOTTSVILLE 138kV	Rebuild line with higher-capacity conductor	\$8,109,000	99% (\$8,043,000)
YANKEE DOODLE / RED HILL 138kV	Rebuild line with higher-capacity conductor	\$3,328,000	99% (\$3,300,000)
BLUFF CITY / REACH 138kV line	Rebuild line with higher-capacity conductor	\$44,563,000	92% (\$41,215,000)
CHARLOTTESVILLE / MOHICAN 138kV	Rebuild line with higher-capacity conductor	\$2,240,000	86% (\$1,917,000)
AMBER / BREMO BLUFF 138kV	Rebuild line with higher-capacity conductor	\$12,600,000	93% (\$11,692,000)
BARRACKS RD SUBSTATION / MOHICAN 138kV line	Rebuild line with higher-capacity conductor	\$8,063,000	86% (\$6,899,000)
BARRACKS ROAD / TIMKEN MERCY SUBSTATION 138kV Line	Rebuild line with higher-capacity conductor	\$9,410,000	86% (\$8,052,000)
SCOTTSVILLE / REACH 138kV L	Reconductor line	\$3,504,000	100% (\$3,504,000)
WALWORTH SUBSTATION / AMBER 138kV L	Reconductor line	\$7,805,000	100% (\$7,805,000)
	Total allocated to project		\$101,272,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.